

STORING DATA ON I²C EEPROM USING AVR MICROCONTROLLER

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Microcontroller-based applications like robotics are often limited by the small storage space of microcontrollers. Implementing applications like artificial intelligence requires a large storage space for pictures, music or text needed. An electrically erasable programmable read-only memory (EEPROM) device, however, may be added to the project as an external storage device.

Here we describe how to store data in an AT24C64 EEPROM and verify the stored data using an ATmega2560 AVR microcontroller.

AT24C64 I²C EEPROM

The AT24C64 is an inter-integrated-circuit (I²C) EEPROM device. The I²C interface is termed as a two-wire

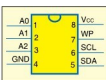


Fig. 1: Pin details of AT24C64 I²C device

interface (TWI) in the context of AVR microcontrollers. I²C is a synchronous serial protocol that allows interconnection of up to 128 different devices using only two bidirectional bus wires: clock (SCL) and data (SDA).

Pin details of an AT24C64 chip are shown in Fig. 1. When using only one device, connect address pins A0, A1 and A2 to ground. The pins form different addresses when SDA and SCL are connected to more than one I²C devices (maximum eight devices) in parallel. For a single device, permanently ground WP pin to enable normal write operations. All the devices connected to the bus have individual addresses, and mechanisms for resolving bus contention are inherent in the TWI protocol.

For the algorithm for handling the TWI interface of the AVR microcontroller, you may refer the datasheet which supports TWI communication. This article covers implementation of the TWI communication protocol in



AT24C64 with ATmega2560 chip.

ATmega2560 microcontroller

The ATmega2560 from ATMEL is a low-power, 8-bit CMOS microcontroller based on the AVR enhanced RISC architecture. The AVR core combines a rich instruction set with 32 general-purpose working registers. The microcontroller has 256 kB of in-system programmable Flash with read-while-write capabilities, 4kB EEPROM, 8kB SRAM, 86 general-purpose input/output (I/O) lines, real-time counter, six flexible timers/counters with compare modes and pulse-width modulation (PWM), four USARTs, a byte-oriented two-wire serial interface, a 16-channel, 10-bit analogue-to-digital converter (ADC) with optional differential input stage with programmable

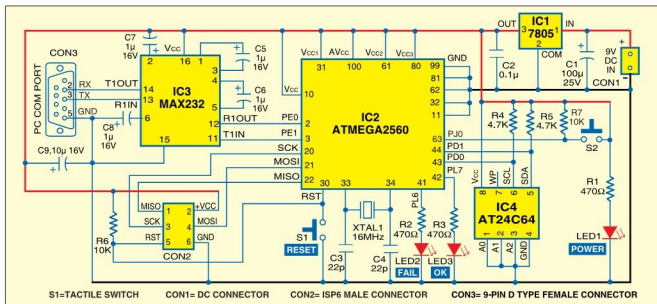


Fig. 2: Storing data on I²C EEPROM using AVR microcontroller